NeuroMPS: A Provenance-Tracked MEA Data Asset and a Calibrated Two-Compartment Digital Twin

Technical report. *Entry for the AI4S Open Innovation: AI + Organ-on-a-Chip challenge (5th Pazhou Algorithm Competition).*

> ***Research decision-support only.*** *Nothing here is an efficacy, safety, diagnostic or clinical claim, and nothing here should be used to make one.*

Submission category: End-to-End System. We built a complete path from public recordings to actionable output: provenance-enforced ingestion, a standardized feature layer, models, a calibrated digital twin, and a harness that tells us which measurements the data can and cannot support. An interactive demo sits on top.

Team. We combine health science with mathematics and machine learning.

member	role	background
Yi Nuo Zhao	Team Leader	Health science student
Ivan T. Ivanov	Machine Learning Specialist	Professor of Mathematics and ML/AI developer

Code and demo	https://github.com/Vodolay/AI4S; the interactive demo runs locally with one command (§21)
Video	https://www.youtube.com/watch?v=WnnD-oSFdcI, a walkthrough of the project and the demo
Verification harness	`python run_all.py --quick` runs the complete pipeline, models and metrics
Audit discipline	`python -m neuromps.report.verify` checks every number in this report against `outputs/metrics.json`
Regulatory position	Research decision-support only. We make no clinical, diagnostic, efficacy or safety claim.

You can reproduce and check everything in this report. One command, `python run_all.py --quick`, runs every stage of the pipeline in about nine minutes on our 20-core workstation (Appendix K), and `python -m neuromps.report.verify` re-checks every number below against the output that produced it. In the source of this report, each computed number carries a hidden pointer to the `outputs/metrics.json` key it came from, so a stale figure cannot quietly survive a rerun. We kept a record of everything we tried and abandoned in `report/LAB_NOTEBOOK.md`.

Executive summary

Neural microelectrode arrays (MEAs) are one of the most useful readouts in organ-on-a-chip work: you grow neurons on a grid of electrodes and watch how their firing changes when you add a compound. The trouble starts when you try to compare results across laboratories. Every group computes its features with its own software and its own definitions, and most deposit summary tables instead of raw spikes, so a number in one paper rarely means the same thing as the "same" number in another.

We built NeuroMPS to tackle that problem directly, and to find out honestly how far the data can be pushed. It has four parts, and after the first round we added a fifth.

- **A standardized data asset.** We assembled 21,887 records from eight public sources, each verified against a SHA-256 checksum of its original file. One feature extractor computes every number from the 1,512 spike-time recordings, so comparisons inside the asset are exact. When we ran the depositors' own meaRtools pipeline on the same recordings, it disagreed

with ours by more than a factor of two on burst rate (§3) — exactly the kind of divergence the asset exists to remove.

- **Benchmarks on real recordings.** We grouped every cross-validation split by culture and bootstrapped over cultures rather than rows. Under those rules, within one laboratory, a simple linear model (balanced accuracy 0.537) beats gradient boosting (0.398), and a model trained only on rat recordings still scores 0.55 on human iPSC recordings. Across laboratories, though, the same classifier falls to 0.3482 against a chance of 0.3333: what it learned in one laboratory did not carry to the next. Two rate features alone match the full nine-feature vector.
- **A biophysical digital twin.** We simulate an asymmetric two-compartment chip in Brian2: leaky integrate-and-fire neurons, 80/20 excitatory/inhibitory, with spike-frequency adaptation, wired through a directional "axon diode". We calibrated it against the interquartile band of 48 real chamber recordings and score it with the same feature code as the real data. We use it for experimental power analysis and as a generator of labelled training data: 3,000 simulated recordings, each with known parameters, train a neural posterior estimator, a masked autoregressive flow, that returns a full posterior for a new recording in 0.31 s without running the simulator again.
- **Negative results and practical guidance.** Nine of our findings set firm boundaries, and §18 pairs each with what we would do instead. For example, chamber-level readouts cannot tell which way axons are firing without electrodes inside the microchannels (0.3854 against 0.5 chance), and the twin's inverse model misses a bicuculline dose trend that two burst-rate features carry. In §20 we turn these findings into recommendations for chip designers, assay developers and analysis pipelines.
- **From chip to animal (Act VII, added after the first round).** We applied the twin's drug knobs to a simulated whole adult fly brain built from its mapped wiring, and read it both as a chip and through the neurons that drive feeding and grooming, with every prediction registered before the runs. The chip's firing rate ranks drugs by how much they disrupt behaviour (Spearman 0.863), but in every drug family behaviour failed at a lower dose than the chip flagged; GABA block had moved firing rate by only 0.149 log2 when the brain's bitter gate on feeding had already broken. Electrodes on the brain's output tract, the fly's counterpart of a chip's microchannels, caught it in time where a planar array did not. A second, male connectome replicates the behaviour and ranking results, and read at its motor neurons the chip is never earlier than motor output.

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Appendices: consolidated technical reference	Appendix A & B — Record architecture, data levels and ingestion ETL Appendix C — Data sources and licenses Appendix D & E — The nine-feature vector, and enforced repository invariants Appendix F & G — Coupling asymmetry proof, and posterior calibration diagnostics Appendix H — Quality-control flag registry Appendix I — Digital twin forward-model parameters Appendix J & K — Microchannel detector robustness, and timings Appendix L, M & N — Sensory validation, dose tracking and species transfer Appendix O — Third-party software, models and AI tools

ACT I: Foundations and the data asset

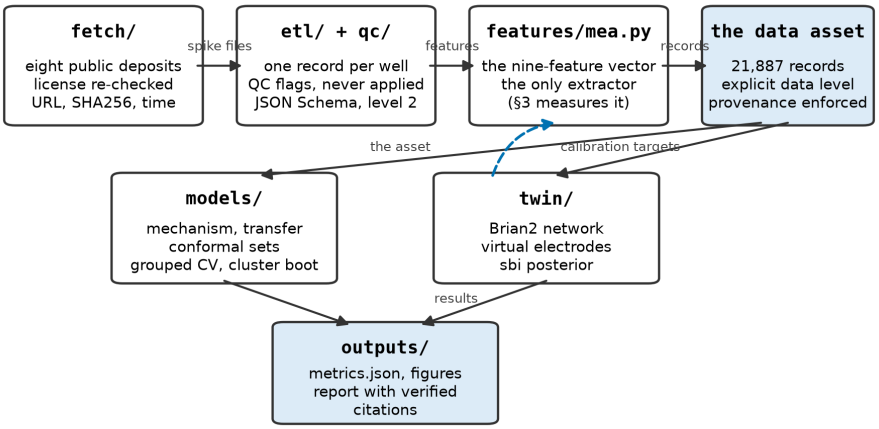
1. What this is, and what it is not

A microelectrode array is a grid of tiny listening posts under a living neural culture. Neurobiologists use it to answer a simple question: does this compound change how the network fires, and if so, how? In practice, every laboratory answers it differently. Labs use different plates, different spike sorters and different definitions of what counts as a burst, so two papers can study the same compound on the same cell type and still not be comparable. On top of that, most groups deposit only a table of summary numbers rather than the spikes themselves — a bit like receiving the book review instead of the book.

Where this matters. The people who feel this most are the ones who need to combine data they did not record themselves: a lab checking whether a compound behaves on its own platform the way it did in a published study, a contract research organization deciding which features to report, or a chip designer trying to prove that a microchannel really makes signals flow one way. We built NeuroMPS for exactly those situations.

We tackle the problem in three steps, and this report follows the same order.

- **We build one harmonized dataset.** We assembled eight public sources into one table, with one exact definition for every feature. A number from Kapucu's plate and a number from ScreenNeuroPharm's chamber are comparable because the same code computed both from the same kind of raw spike input. Every record says where it came from and under what license, and carries a checksum of its original file.
- **We ask what that dataset can support.** We run models on the real recordings to find out which conclusions survive a change of laboratory, of species and of platform — and which questions a chamber-level readout cannot answer at all.
- **We build a digital twin for the questions the recordings cannot answer.** It is a forward simulation of a two-compartment neural chip: two populations of model neurons wired through a directional microchannel pathway, with virtual electrodes over them. We tune it against real chamber baselines from the dataset and score it with the same feature code, so it cannot make itself look good by being measured differently.



Provenance is enforced at the asset boundary: the build refuses to emit an asset if any record's hash disagrees with the download manifest.

Figure 1. *System architecture. REAL recordings and SIMULATED twin output pass through the same feature extractor, and provenance is enforced at the boundary of the data asset.*

What it is not. NeuroMPS is not an automated screening platform, and it does not predict whether an uncharacterized compound is safe or effective. Several of our most useful results are negative: they show that a measurement does not contain a signal people routinely assume it does. We treat those results (§18) as a core deliverable, not as an apology.

2. The data

Our asset holds 21,887 records from eight public sources. They are not all the same kind of record, and that difference matters more than the count.

what the record holds	record count	what you can do with it
spike times	1,512	recompute any feature from scratch with our common code
fitted dose-response curves	20,375	read a potency estimate that another laboratory fitted

We require every record in our dataset to explicitly declare its data type. While that might sound like routine bookkeeping, it directly protects our pipeline: virtually all of our predictive modeling and twin calibration rely solely on spike-time recordings, because only those can pass through our unified feature extractor. If we had simply lumped the two record types together into a single table, the dataset would look twenty times larger — but it would be twenty times less useful to us.

The species imbalance. By species, our asset holds 21,530 rat and 357 human records. This imbalance is the single biggest limitation of the project, and no amount of clever coding can fix it: it reflects what is publicly available under licenses that permit reuse. Rather than hide it, we treat it as an explicit boundary whenever we test species transfer.

The case that tested our rules. One dataset (Tukker 2020) was an ideal scientific fit, and we still excluded it. It is licensed CC BY-NC, and the competition rules give the organizing committee joint ownership of entries — a right that a non-commercial license does not allow us to grant. We record the exclusion and its reason in `data/raw/MANIFEST.json`, next to the sources we accepted, instead of letting the dataset quietly disappear. We mention it because you should be able to check that our license discipline is real rather than claimed: every source re-checks its license at fetch time, and the build stops if any checksum disagrees with the manifest.

ACT II: Standardizing the mess and benchmarking real biology

3. Are two laboratories' numbers actually comparable?

The whole dataset rests on this question, so we tested it instead of assuming it. One module, `features/mea.py`, computes all nine features directly from spike times: mean firing rate, burst rate, burst duration, spikes per burst, percentage of spikes in bursts, network burst rate, synchrony index, the coefficient of variation of inter-spike intervals, and the number of active electrodes.

We detect bursts with a stated rule — spikes closer than 100 ms, at least five in a row — rather than a vendor default. For synchrony we use the spike-time tiling coefficient (STTC) with a 50 ms coincidence window. Unlike a plain correlation, STTC does not reward two electrodes just for both being busy.

The benchmark. Kapucu and colleagues deposited their own feature tables alongside their spike files, computed with `meaRtools`. That gave us an independent implementation to test against: we ran our code over their spikes and compared the results across 336 recordings on 3 plates, 2,298 paired values in total. We consider this the most important methodological result in the report, so here it is, plainly.

feature	rank agreement (ρ)	value ratio (ours / theirs)	practical consequence
`synchrony_index`	0.9839	0.736	ranks match; our STTC is about a quarter lower
`mfr_hz`	0.9831	0.7539	ranks match; baseline firing rate scaled differently
`active_electrodes`	0.9763	1.3624	ranks match; our activity cutoff is more inclusive
`pct_spikes_in_bursts`	0.9703	0.9864	reproduces on both axes
`spikes_per_burst`	0.9167	0.9418	close numerical agreement
`burst_rate_per_min`	0.8907	0.4423	disagrees twofold, purely from definition
`burst_duration_s`	0.7926	0.8189	the weakest rank agreement of the seven

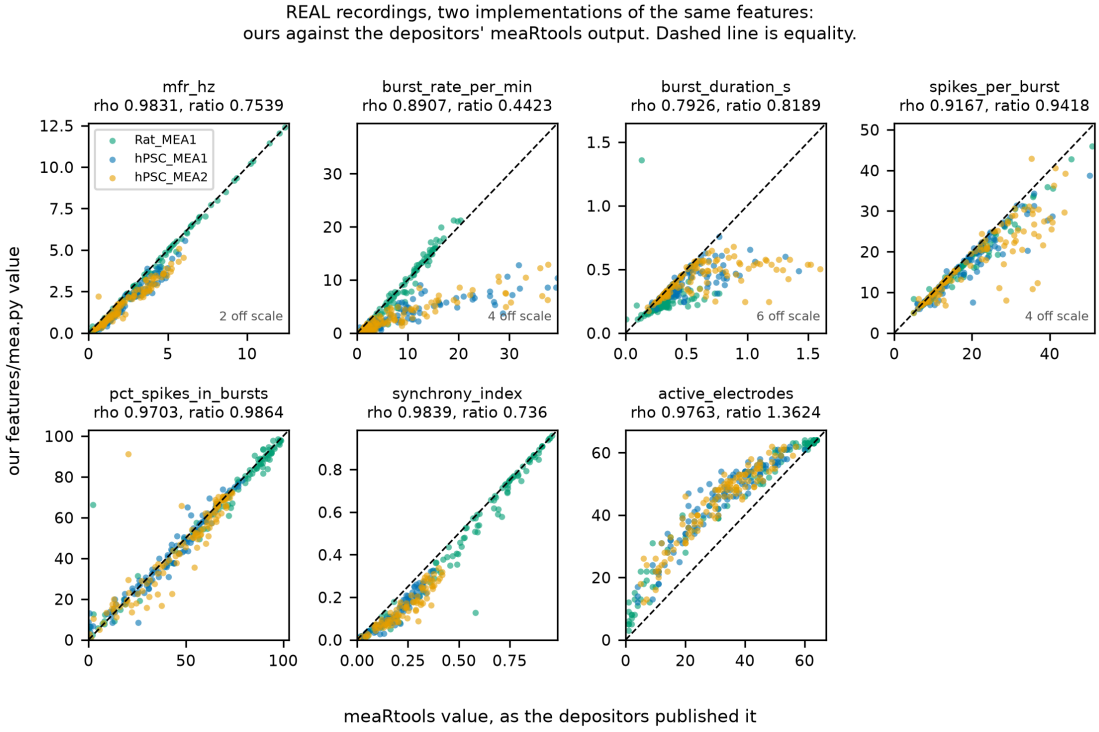


Figure 2. *Our extractor against the depositors' own meaRtools output, REAL recordings. Rank agreement on the horizontal axis, ratio of values on the vertical. Only `pct_spikes_in_bursts` lands on equality; burst rate differs twofold from definitional choice alone.*

Read the two columns separately. The left one asks "do we rank recordings the same way?", and the right one asks "do we get the same number?" Only one feature passes both tests. Three rank recordings almost identically but on a different scale. And the burst measures disagree about *which* recordings burst more: `burst_rate_per_min` differs by more than a factor of two.

What this means. Inside our dataset, comparisons are safe, because one code path produced every number. Across papers they are not: "bursts per minute" from two laboratories can differ twofold

from the algorithm definitions alone, before any biology enters the picture. That is exactly why we recompute everything from raw spikes instead of pooling published summary tables.

4. The three questions we asked the data

Every model in this report works on changes, not on static states. We pair each treated well with its own pre-treatment baseline, and the model sees the log2 fold-change of each feature between the two. This is the most consequential design decision in our analysis, and we made it for the same reason an experimenter would: cultures differ enormously from one another. If one culture fires at 4 Hz and another at 8 Hz, that tells you about the cultures. If a culture drops from 8 Hz to 4 Hz after a compound goes in, that is a pharmacological response.

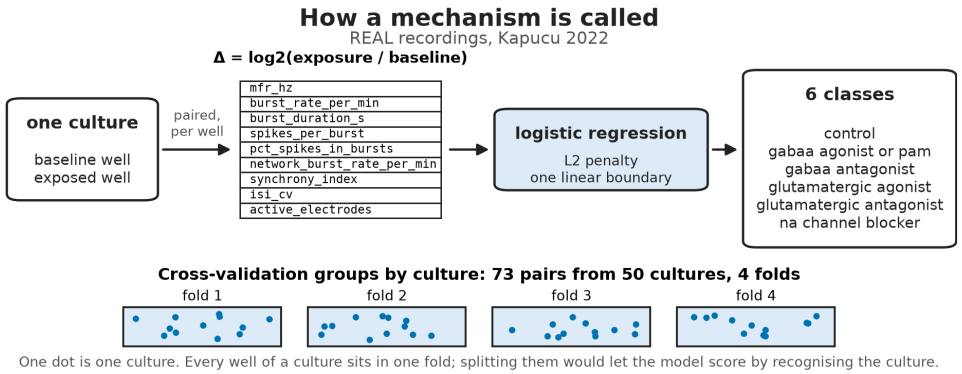
For each feature, measured in the same well before and during exposure, the model sees

$$\Delta = \log_2(x_{\text{exposure}} / x_{\text{baseline}})$$

so every halving or doubling moves Δ by one unit, whatever the culture's starting rate. A well that falls silent is floored at one twentieth of its baseline, so abolished activity stays a finite number, and a zero baseline leaves the ratio undefined rather than inventing a value for it.

1. **Mechanism.** Given the change in the nine features, can we tell which class of compound caused it — GABA-A antagonist, sodium-channel blocker, glutamatergic antagonist, and so on?
2. **Detection.** Can a model trained on one laboratory's data flag a known compound in a different laboratory's recordings?
3. **Platform contrast.** Does the same mechanism look the same on two different platforms?

Two rules we never break, and they are why our scores look lower than many published ones. First, *we split strictly by culture*: wells from the same culture never end up on both sides of a training/testing split. Wells within a culture are near-copies of each other, so splitting them would let a model score well just by recognizing the culture — like seeing the exam questions in advance. Second, *we resample uncertainty over cultures, not rows*: every interval is a cluster bootstrap over whole cultures. With 73 pairs from 50 cultures, treating each row as independent would produce intervals about half as wide — and wrong.



Balanced accuracy 0.537 on [0.4216, 0.6845], against a chance of 0.167. The interval is wide because the study is small, and it is reported that way.

Figure 3. *How a mechanism call is made, on REAL recordings: each culture's paired wells become a nine-feature log-ratio vector, one regularized logistic regression maps it to a mechanism class, and cross-validation keeps every culture inside a single fold.*

5. Can you read a mechanism from the features?

Within one platform, we worked with 73 treated/baseline pairs from 50 cultures across six mechanism classes (Kapucu 2022):

model	balanced accuracy	95% bootstrap CI	operational note
always guess commonest class	0.167	—	theoretical chance across 6 classes
gradient boosting	0.398	0.2762 to 0.5178	expressive model over-fits 73 examples
logistic regression (L2)	**0.537**	0.4216 to 0.6845	simpler linear boundary generalizes best

Balanced accuracy averages the recall of each class, so a model cannot win just by getting the most common class right. Three things stand out. First, there is real signal: the model calls roughly half the mechanisms correctly where chance is 0.167, and the interval clears chance. Second, the interval is wide, because 73 pairs from 50 cultures is a small study, and we show that openly. Third, the simpler model wins decisively. When a flexible model loses to a linear one by fourteen points, the honest reading is that our dataset is small, not that boosting is bad.

The errors are not random either. They cluster between the two GABA-A classes, antagonists and positive modulators, which makes chemical sense given how both act on the excitatory/inhibitory balance.

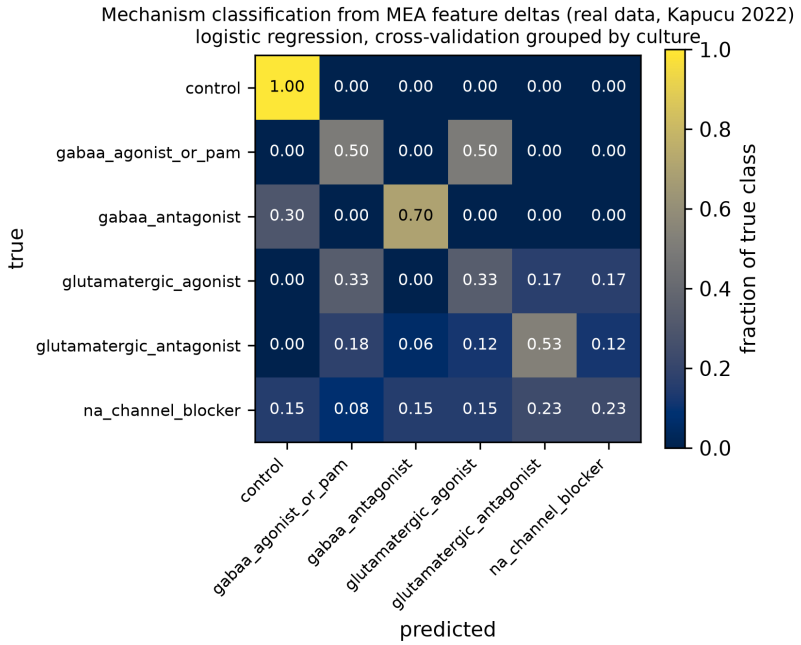


Figure 4. *Task 1 confusion matrix, REAL data: out-of-fold predictions over 73 treated/baseline pairs from grouped cross-validation. Errors concentrate between the two GABA-A classes.*

The same task, refit in a second laboratory. `vassallo2017` comes from a different laboratory and a different vendor, with six compounds Kapucu never used, and each baseline sits on the same chip as its doses: 171 pairs over 26 chips in four classes. We refit the same pipeline there from scratch, training and testing inside that source with the same grouped cross-validation. Logistic regression reaches 0.333 against a chance of 0.25, on an interval of 0.2317 to 0.4204 that includes chance.

These two results are not a before and after. They are different problems — six classes against four, different compounds, and therefore different chance levels — so the only fair comparison is each against its own baseline. Kapucu clears its chance of 0.167 more than threefold; Vassallo clears its own by about a third, and its interval does not exclude chance. So mechanism signal is not a quirk

of the one dataset we built the method around, but it is much weaker in the second laboratory. Note that nothing is transferred here: both models are trained and tested inside their own laboratory. We test transfer itself — a model trained in one laboratory and applied to another — in §6 and at the end of this section.

Why we never pooled the platforms. Pooling is tempting, and it would have produced a better-looking number. It would also have been meaningless. Across the full pool, the label alone determines the source for 97.35% of pairs, and 6 of the 9 mechanism classes appear in only one source. A pooled classifier would score well simply by learning which laboratory a recording came from.

The structural reason, and a test built around it. The obstacle is even sharper than that percentage suggests: the largest number of classes that any two laboratories share is 1, and a classifier needs at least two classes to tell apart. That means we cannot run leave-one-laboratory-out cross-validation here at all.

What we can build is a deliberately confounded test. Three classes appear in more than one source, and all three appear in Kapucu. We train there on 20 pairs and test on the other laboratories' versions of the same three classes, 535 pairs. In that test set, each class comes from exactly one laboratory — a correspondence of 1.0 — so in this test, class and laboratory cannot be told apart. A high score would have been ambiguous: it could mean the model read the pharmacology, or only that each laboratory's recordings look different. A score at chance has one reading: neither the mechanism signal learned in Kapucu nor any laboratory difference carried the classes across. That is why we built the test so that only a **low** score is interpretable.

The model scores 0.3482, 95% CI [0.278, 0.4082], against a chance of 0.3333. That is indistinguishable from guessing: what the model learned in one laboratory did not carry to the others.

The control that makes it mean something. Twenty training pairs is not many, so the failure could come from sample size rather than from the change of laboratory. To check, we scored the same classes and the same 20 pairs *inside* Kapucu with grouped cross-validation, which gives 0.5833. Same data, same size: the model learns something when it stays in one laboratory and nothing when it crosses into another.

6. External detection, and the negative control that matters more

Can a model trained on Kapucu's plate flag bicuculline in ScreenNeuroPharm's chambers — a different laboratory, and a different compound from the same mechanism class? Across 497 recordings, it calls "GABA-A antagonist" 36.6% of the time. There are two ways to read that, and only one is right. It is far above the rate at which that class would appear by chance, so something does transfer. But it is not an autonomous detector, because two-thirds of the time it says something else.

The negative control. What makes that first number trustworthy is the control. On Sharf's diazepam recordings — a GABA-A positive modulator, the opposite pharmacology — the same model calls "antagonist" only 7.7% of the time. A model that simply shouted "GABA-A antagonist" at anything unusual would have fired on both. Ours does not.

Detection does not track dose. Within the bicuculline series, detection probability shows no relationship with dose: Spearman $\rho = -0.059$, $p = 0.189$. Mechanism and dose are different questions, and this model only answers the first. The recordings do carry a dose signal, as §18 shows; a classifier built to tell mechanisms apart does not read it.

7. Rat to human

The whole promise of organ-on-a-chip work is that a human preparation tells you something a rodent one cannot. So we trained on rat and tested on human, using only the five mechanism classes present in both.

transfer setting	balanced accuracy	chance	empirical finding
rat, tested within rat	0.543	0.2	rodent primary culture baseline
human, tested within human	0.646	0.2	human iPSC model evaluation
rat → human, no adaptation	**0.55**	0.2	transfers without loss; permutation p = 0.0002
rat → human, CORAL	0.442	0.2	covariance alignment erodes biological signal
rat → human, standardized	0.408	0.2	per-species normalization degrades toward chance

The transfer is real, not luck, and two things stand out. First, rat → human (0.55) is *not* worse than rat tested within rat (0.543): mechanism information crosses the species boundary about as well as it crosses between cultures within rat. Second, **both domain-adaptation methods made it worse.** These are the standard tools for this problem, but here they removed real signal rather than distribution shift — which is what happens when a method designed for thousands of samples meets 37 rat training pairs and 32 human testing pairs. We report the unadapted number as our result, with the adapted ones beside it, rather than picking whichever was highest.

8. Which features are actually doing the work?

If the nine-feature vector were doing nine distinct jobs, dropping most of it should hurt. It does not.

evaluated feature subset	balanced accuracy	included features
the two rate features only	**0.53**	`mfr_hz`, `burst_rate_per_min`
species-invariant subset	0.516	features stable across untreated species baselines
all nine features	0.514	the full biophysical complement
synchrony only	0.455	`synchrony_index` (STTC)
firing rate only	0.442	`mfr_hz`
network features only	0.417	`network_burst_rate_per_min`, `synchrony_index`
burst features only	0.411	`burst_duration_s`, `spikes_per_burst`, `pct_spikes_in_bursts`

The two rate features alone do at least as well as all nine. The intervals overlap heavily, so we do not claim that rate-only is better; what we can say is that the other seven features add nothing we can detect at this sample size. That matters in practice, because the burst and synchrony measures — the very features that differ most between laboratories (§3) — are also the ones carrying the least mechanism information here.

Validation on sensory neurons. Because rate alone looked sufficient, we could bring in a source that reports only firing rates: `fofie2025`, human iPSC-derived sensory neurons, 291 pairs from 48 wells. Training on Kapucu's cortical rate changes and testing on these sensory neurons gives 0.5012 against a chance of 0.3333, using a single feature and a preparation with a completely different neuronal identity. What the two sources share is the compound's mechanism, not the preparation or the network.

9. How confident is the model, really?

Calibration. When the model says 70%, is it right 70% of the time? Our expected calibration error is 0.1114 on 73 points, with a bootstrap interval reaching 0.2557. That upper end is large. We would

treat the predicted probabilities as a ranking, not as a risk — and not even as a reliable ranking above 0.8 confidence, where observed accuracy actually falls.

Conformal sets. Instead of a single answer, a conformal predictor returns a set of classes that is guaranteed to contain the truth a stated fraction of the time. When we ask for 95%, empirical coverage is 0.918 and the average set holds 4.562 of the six classes. A set of four or five classes out of six is honest and nearly useless — which is the right answer for 73 calibration points. A method that promised 95% coverage with narrow sets on this much data would be doing something wrong.

10. Untreated baselines: what differs before a drug is added

Let's set pharmacology aside for a moment. If we take untreated recordings and ask what separates rat from human, and dissociated cultures from organoids, the features consistently fall into two groups across three separate analyses. Rate-like features stay fairly stable across species, while synchrony and burst-structure features move the most. Those same features differ most between baselines, carry the least mechanism information (§8), and are the least reproducible across software (§3).

That three-way agreement is the most actionable result here for experimental design: the burst and synchrony family is where you will have the most trouble, from three independent directions at once.

One analysis disagrees, and we kept it. The platform contrast (§11) finds that for the *same* mechanism on two platforms, mean firing rate reverses sign while synchrony agrees. So three analyses say synchrony is the unreliable family, and one says it is the reliable one. We do not resolve that tension; we report it.

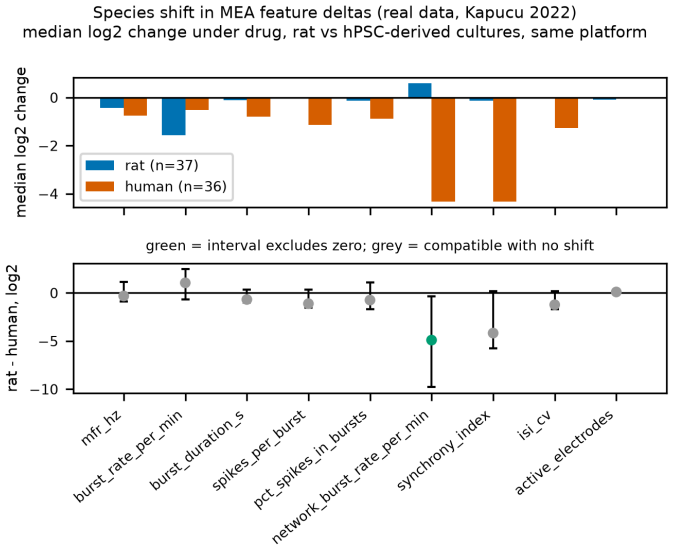


Figure 5. *Per-feature species shift, REAL data: rat against human baseline distributions with cluster-bootstrap intervals. Rate and burst-fraction features move least; synchrony moves most.*

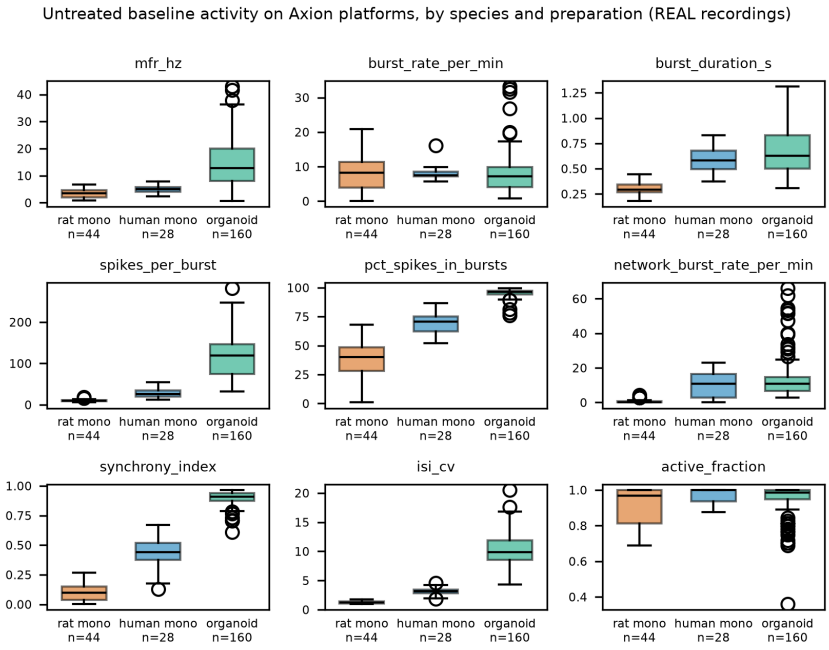


Figure 6. *Untreated baseline activity, REAL recordings, by species and preparation. Burst rate is the one feature flat across all three groups.*

11. The cross-platform reversal paradox

We compared GABA-A antagonists at 30 µM and above on two platforms: Kapucu's gabazine and ScreenNeuroPharm's bicuculline. Mean firing rate goes **up** on one platform and **down** on the other, while synchrony moves the same way on both.

If you take away one thing from this report about cross-laboratory comparison, make it this: the mechanism is identical, yet the direction of the most commonly reported endpoint is not. Anyone pooling MEA pharmacology on the assumption that "GABA-A antagonists raise firing rate" would get it backwards half the time.

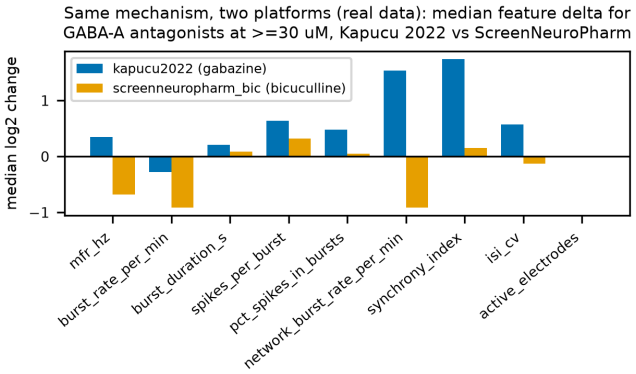


Figure 7. *The same mechanism on two platforms, REAL data: per-feature log2 change for GABA-A antagonists. Mean firing rate reverses sign across platforms; synchrony agrees.*

ACT III: Calibrating a biophysical digital twin

12. The digital twin: what it is, and what it is for

Our digital twin is a simulated culture, not a statistical surrogate. It has two compartments of model neurons — 80% excitatory, 20% inhibitory, each a leaky integrate-and-fire neuron with spike-frequency adaptation — wired densely within each compartment and sparsely between them through a directional pathway. Virtual electrodes sit over the sheet and "hear" neurons within a set radius, and their spike trains go through the same `features/mea.py` that the real recordings use. That last point is a deliberate constraint: the twin cannot look good by being measured differently.

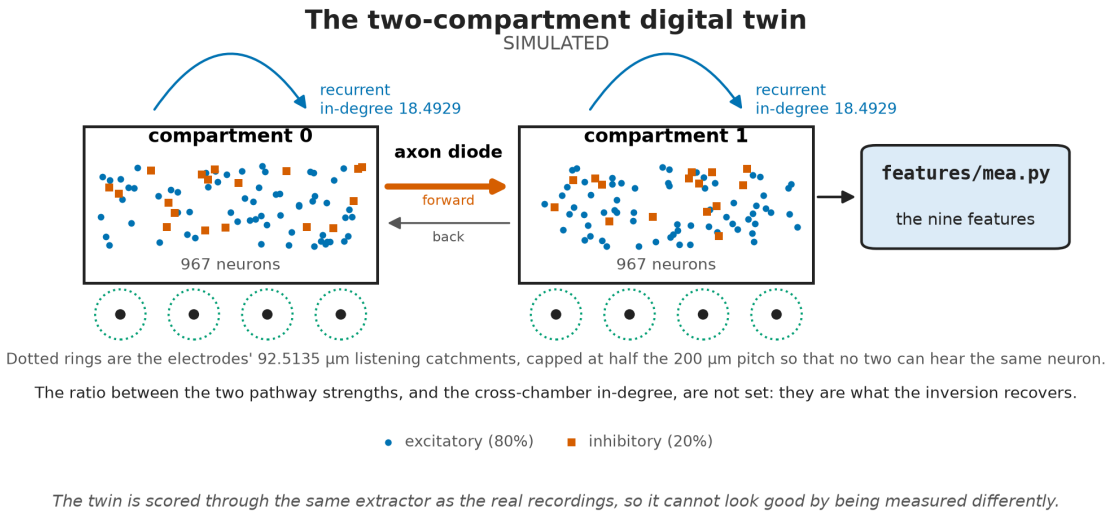


Figure 8. *The SIMULATED two-compartment twin: two chambers of model neurons, a one-way axon diode between them, and virtual electrodes whose listening radius is capped so that no two can hear the same neuron.*

What the twin is for. It lets us test scenarios that are expensive or impractical to try at the bench. How many electrodes, and how much recording time, do you need to detect an effect? What happens to the readout if axonal projections become twice as directional? And, most fundamentally, is the metric you are tracking even capable of answering your question?

What calibration means here. We are not fitting the twin to reproduce one culture — that would simply be overfitting to that culture. Instead, we ask it to land inside the interquartile band of 48 real chamber baselines from rat two-chamber devices. In other words, we want it to be an unremarkable member of the population, not a copy of one member.

13. Calibration

We ran a random search over 9 parameters, scoring each candidate on 2 network seeds across 160 draws. None of the chosen values sits on the edge of the range it was drawn from, so the optimum is genuinely inside the search space rather than clipped at a boundary.

feature	simulated twin value	real band (IQR)	calibration result
`synchrony_index` (STTC)	0.8387	0.678 to 0.843	inside target band
`burst_rate_per_min`	22.9219	12.627 to 29.726	inside target band
`network_burst_rate_per_min`	24.0	11.692 to 22.697	minor overshoot
`mfr_hz`	5.9862	6.793 to 21.007	minor undershoot
`pct_spikes_in_bursts`	83.8409	84.657 to 93.685	near lower bound

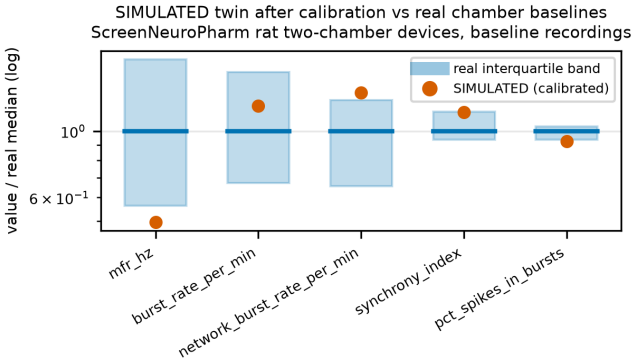


Figure 9. *SIMULATED against REAL: the calibrated twin's baseline features against the real baseline band from rat two-chamber devices.*

Getting to this table took us longer than anything else, and the story is a warning for anyone who simulates cultures.

For a long time the twin was far too synchronous — an STTC around 0.96 against a real band topping out at 0.84 — and it looked as if we were missing a biophysical mechanism. We tried three candidates, and none of them worked: short-term synaptic depression, varied axonal conduction delays, and distance-dependent wiring, which actually pushed synchrony the wrong way.

The real cause was not biophysics; it was cell density and electrode geometry. Our simulated chamber held 250 neurons in 1.92 mm², about 130 neurons/mm², whereas a real dissociated culture runs roughly 1000 to 3000. On top of that, we had let the electrode's listening radius grow larger than the spacing between electrodes. The oversized electrode compensated for the sparse culture in two ways at once: it pooled enough neurons to produce a plausible firing rate, and, because neighboring listening areas overlapped, it counted one neuron's spikes on several electrodes — which inflates pairwise synchrony by construction, with no biology involved. Once we fixed the density at 967 neurons per compartment and imposed one geometric rule, synchrony landed inside the real band for the first time:

An electrode's listening radius may not exceed half the electrode pitch, r ≤ p/2.

On our 200 µm grid, that rule guarantees that no neuron is heard by two electrodes. The calibrated radius settled at 92.5135 µm, inside the cap. The rule carries over to physical arrays: if an electrode's catchment exceeds half the pitch, expect synchrony to read high for reasons that have nothing to do with the culture.

Out-of-sample trust. The winning parameter set reached a loss of 0.0756 on its selection seeds. On 5 seeds the search never saw, it scores 0.1994, with synchrony holding at 0.7732 ± 0.0685 and inside the band on 60% of seeds. Just like a real culture, the twin varies enough from run to run that a single simulated recording can fall outside the band.

14. Drug knobs: does the twin respond like a culture?

The twin has three pharmacological knobs: it can scale inhibitory transmission (`gabaa_scale`), scale excitatory transmission (`ampa_scale`), or raise the spike threshold (`threshold_scale`). We tested directional validity — does each knob move each feature in the right direction? — and we only scored features whose real direction is established by a cluster bootstrap.

pharmacological knob	mechanistic action	directional concordance
AMPA blockade	downscale excitation	4 / 4
sodium-channel block	elevate spike threshold	3 / 3
GABA-A antagonism	downscale inhibition	4 / 5
GABA-A potentiation	upscale inhibition	1 / 4

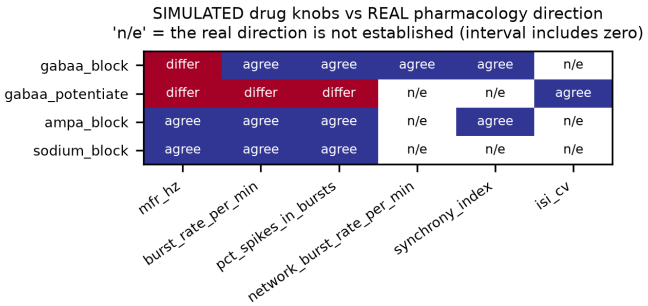


Figure 10. *SIMULATED drug knobs against REAL pharmacology directions, per feature. Only features whose real direction is established by a cluster bootstrap are scored.*

Blocking excitation and blocking sodium channels behave correctly on every scored feature. Blocking inhibition is mostly right. Potentiating inhibition is mostly wrong — and that is the knob with the weakest real reference behind it, just 17 pairs from two sources. So the twin is usable for reasoning about strong, well-evidenced perturbations, but we have not validated it for subtle allosteric modulation.

The twin as a source of labelled training data (added after the first round). Directions are one test; recognition is a harder one. We simulated pairs for the five mechanism classes the twin has a knob for (one chip recorded twice, once with the knob moved; 30 pairs per class), trained the §5 classifier on those alone, and applied it to the 73 real Kapucu pairs. Registered before the pairs were simulated (`preregistration/twin_prior_classifier.json`), it reaches a balanced accuracy of 0.41 (95% interval 0.2808 to 0.5072, culture bootstrap) where chance is one in six, without seeing a single real label. It never recognises a sodium-channel blocker, and the reason is size, not direction: the twin's threshold range moves firing by −3.273 log2 and silences most electrodes, where the real blockers move it by −0.795 with every electrode still active. As a prior for the classifier trained on real pairs, the simulated pairs add nothing: balanced accuracy goes from 0.537 to 0.53, and the conformal sets do not shrink. Two of the three registered predictions failed. The twin can stand in for labels where there are none; where real labels exist, it does not improve on them.

ACT IV: Inverting the twin and resolving directionality

15. Two-compartment devices, and a result we did not want

The twin models an asymmetric two-compartment chip — the "axon diode" idea, where microchannel geometry biases which way axons grow. Before simulating it, we asked what real devices actually show.

Tissue identity drives asymmetry. Across 16 devices, we measured how asymmetric the coupling between compartments is. Devices pairing two different tissues (cortex with hippocampus) are more asymmetric, with a median of 0.138, than devices pairing a tissue with itself, median 0.094, p = 0.042. What the chambers contain matters more than the fact that they are connected.

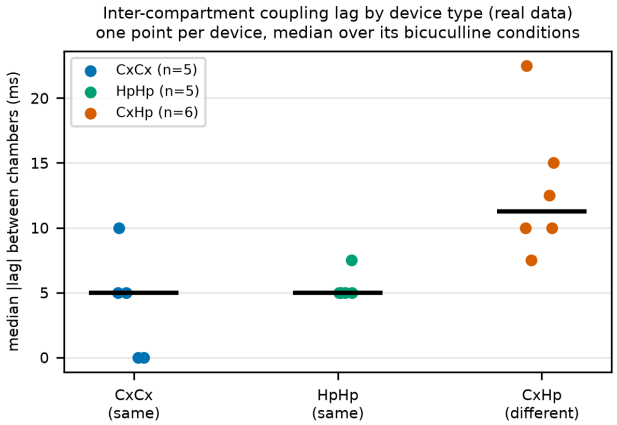


Figure 11. *Inter-compartment coupling lag by device type, REAL data, one value per device: five CxCx, five HpHp, six CxHp. The two same-tissue types are indistinguishable; the heterogeneous type sits apart.*

The directionality null. One deposit recorded axonal crossings directly inside the tunnels, on 10 cultures (Brewer 2021). That let us ask the question a chamber readout exists to answer: if you had only the chamber recordings, could you tell which way the axons were firing? Using all four chamber-level measures together — coupling asymmetry, peak correlation, lag and rate imbalance — a classifier reaches a balanced accuracy of 0.3854 against a chance of 0.5. That is worse than guessing. And it is not just a small-sample failure: with this many cultures, the analysis could have detected a balanced accuracy of 0.6438 if one had been there. We could only recover direction by instrumenting the microchannels themselves.

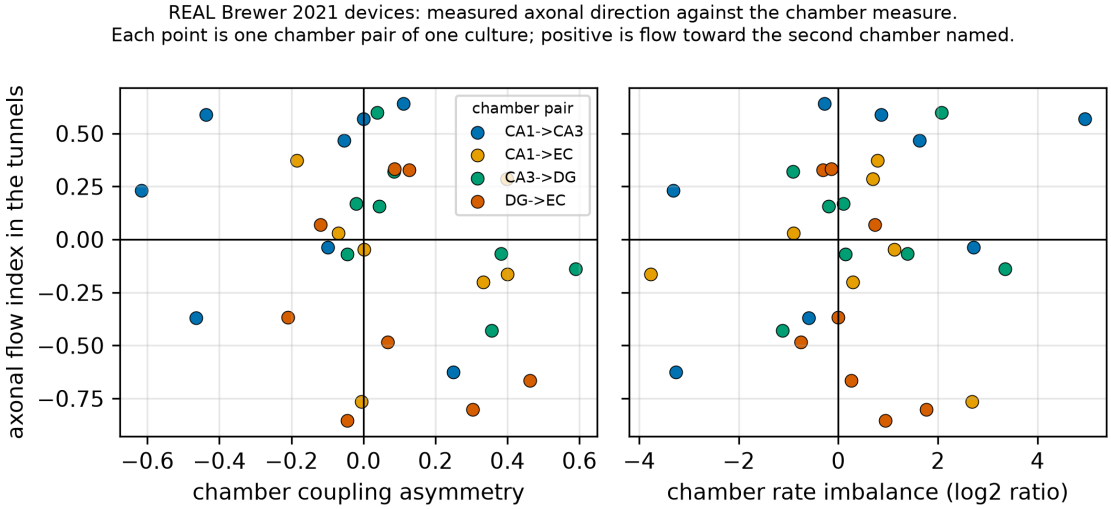


Figure 12. *Measured axonal flow against chamber-level readouts, REAL recordings. The chamber measures do not order the cultures by the direction their axons actually fired.*

16. Running the model backwards, and what that cannot do

The more ambitious use of a twin is to run it backwards: given a recording, infer which knob was turned. We did this with simulation-based inference, using a neural posterior estimator trained on 3,000 simulations at the calibrated cell density. Here the twin works as a generator of labelled data: every simulation is a training pair whose true parameters are known exactly, which no bench experiment can supply.

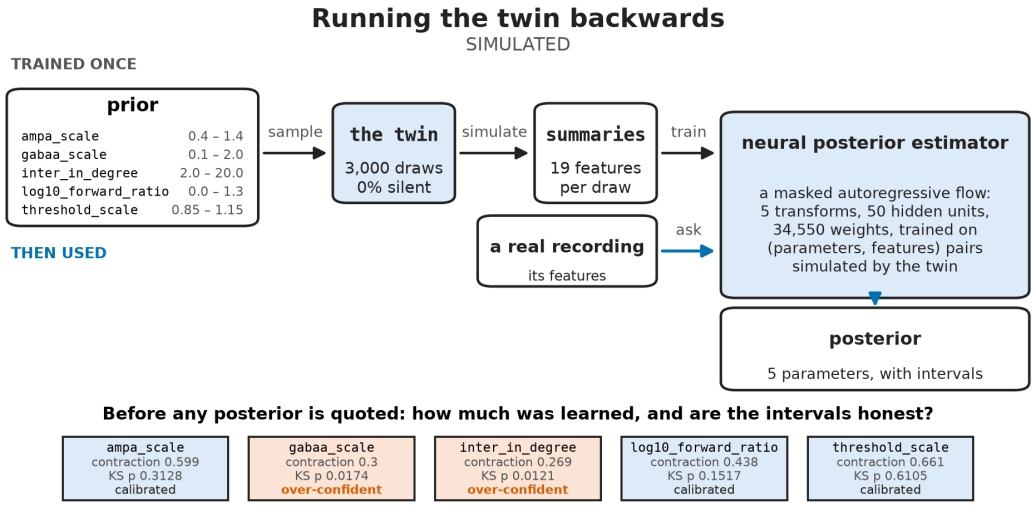


Figure 13. *Running the SIMULATED twin backwards: draws from the prior are simulated and summarized as features, a neural posterior estimator learns to invert them and is then asked about a real recording. The bottom row is the honesty check applied before any posterior is quoted.*

The estimator is a masked autoregressive flow, built with the `sbi` library: a stack of 5 autoregressive transforms with 50 hidden units each, 34,550 trained weights in all. It learns the joint density of the five parameters given the 19 summary features, so it returns a full posterior rather than a point estimate, and it never has to run the simulator again once trained. That is what makes the inference amortized: on 20 held-out simulations, a posterior of 1,000 draws takes a median of 0.3061 s on our workstation, against 6.69 s for one 10-second run of the simulator, which a method that simulates afresh for every recording would need many times over. That speed holds for recordings the twin can produce; for those it cannot, sampling stalls, which is the refusal described below.

The same generator could in principle pre-train models for assays where real data are scarce, such as human iPSC cultures, but only once the twin is calibrated for them. Today it is calibrated on rat devices and sits inside the real band on two of five features, so we have not done this.

Before we trust any posterior, we run two honesty checks.

model parameter	biophysical role	posterior contraction	KS calibration (p)	uniformity status
`threshold_scale`	action-potential threshold	0.661	0.6105	well calibrated
`ampa_scale`	excitatory synaptic weight	0.599	0.3128	well calibrated
`log10_forward_ratio`	axonal pathway asymmetry	0.438	0.1517	well calibrated
`gabaa_scale`	inhibitory synaptic weight	0.3	0.0174	**over-confident**
`inter_in_degree`	cross-chamber connectivity	0.269	0.0121	**over-confident**

Contraction measures how much narrower the posterior is than the prior; 0 means the model learned nothing. Rank uniformity asks whether the stated intervals are honest. Three knobs pass. Inhibitory transmission does not — and it is the knob our pharmacology sections depend on, so **we do not quote its intervals as calibrated.** Before the density correction (§13), no parameter passed and a fifth of our simulations were silent; now 0 are. Simulating at the calibrated density brought three knobs into calibration. Appendix G shows the full rank histograms.

The false-disinhibition artifact. When we applied the model to real untreated chips, it returned an inferred inhibition scale of 0.5219 on an interval of 0.2732 to 0.8503 — apparent disinhibition in wells that had received no drug. That is not biology. It is the model's way of saying "real recordings do not look like my simulations," in the only vocabulary it has. When we reformulated inference around paired deltas, untreated chips moved to 0.9108 on 0.8583 to 0.966, and contraction rose to 0.5375. Most of the offset disappears, but not all of it: the interval still stops short of neutral.

Out-of-distribution rejection. We asked for 200 draws per recording across 113 bicuculline trials, and the posterior could not deliver them for 9 of those recordings; the worst yielded only 1 draw, and 0 failed outright. This only appeared once the twin moved to its calibrated density: a posterior fitted to a sharper forward model is less forgiving of observations that model cannot produce. In effect the posterior carries a built-in out-of-distribution flag: a recording it cannot sample from is one the twin cannot produce, and in high-content screening that recording should go to a person rather than get a forced answer. We have not measured how well this flag separates genuine anomalies from ordinary recordings, so we report it as a behavior, not as a validated detector.

17. What it takes to see directionality at all

One thing a twin is genuinely good at is telling you how much experiment you need. Across 120 simulated networks with known wiring, the cheapest configuration that reaches our target distinguishes asymmetric from symmetric wiring with an AUC of 0.8928, using 24 electrodes per compartment and a 10-second recording.

The hidden catch. Most of that signal is firing rate, not direction. A directional pathway delivers more excitation to its target chamber, so the target simply fires more. If we lower the receiving chamber's background drive, measured coupling asymmetry moves from 0.109 to 0.681 with the wiring unchanged. Across the sweep, drive moves the measure further than wiring does: a span of 0.671 against 0.109. Regress firing rate out of the features, inside each training fold, and the cheapest configuration doubles to 48 electrodes per compartment. Twenty-four electrodes and ten seconds will tell these cultures apart, but claiming that this reflects directional communication requires explicit rate-matching controls.

ACT V: The negative results and scientific boundaries

18. Collection of negative results

Negative results are easy to miss when they are scattered through a long report. Collected in one place, they are the part of our work we would most like to see challenged — and the part most likely to save someone else an experiment. Each one tells you what not to rely on, so the last column says what we would do instead.

#	finding	evidence	what to do instead
1	the twin misses a dose trend the features carry	2 of 7 features fall with dose; inferred inhibition does not (ρ = −0.121)	read dose from burst rates against each device's own baseline; do not use the twin's dose inference until it recovers that trend
2	chamber readouts miss direction	chamber classifier 0.3854 against 0.5 chance	put electrodes inside the microchannels
3	no cross-laboratory transfer	0.3482 against 0.3333 chance	validate inside one laboratory; a cross-laboratory test needs two laboratories sharing at least two classes
4	domain adaptation backfires	CORAL drops transfer to 0.442, standardization to 0.408	report the unadapted transfer first; adapt only once it beats that on held-out cultures
5	synchrony was an artifact	depression and delay dispersion fail to fix synchrony	match cell density to culture and cap each electrode's radius at half the pitch (§13)
6	posterior over-confidence	`gabaa_scale` fails rank uniformity at p = 0.0174	check rank uniformity before quoting an interval; quote only parameters that pass
7	baseline read as drug effect	absolute features infer 0.5219 inhibition scale on untreated chips	infer from paired deltas against each well's own baseline (§16)
8	support rejection	the calibrated twin cannot sample 9 of 113 real recordings	treat a refusal as an out-of-distribution flag and send the recording to a person
9	vesicle depletion does not explain the GABA-A reversal (added after the first round)	under strong GABA-A block the twin's firing rate rises by 2.389 log2; with depletion, by at least 0.8687; on real chips it changes by −0.335	do not read the twin's firing-rate direction under disinhibition as the culture's; look for the missing mechanism elsewhere (recalibrating with depression, depolarization block)

On the first, which matters most. §6 showed that the mechanism classifier had no dose relationship, and §16 showed that the inverse model could not recover dose either. So we asked whether the signal is in the features at all: across 113 bicuculline recordings on 16 devices, does *any* feature move monotonically with dose? Seven of the nine had enough points to test; the two burst-shape features, burst duration and spikes per burst, did not. Using raw values, 2 of the 7 do: burst rate (ρ = −0.58) and network burst rate (ρ = −0.60) fall with dose. After subtracting each device's own baseline, the same 2 do, in both cases after a Holm correction for testing seven features at once. The signal is there, and the twin's posterior does not recover it, so the inverse model is the part to fix. An earlier version of this analysis ranked doses without their units, which put the two lowest doses, recorded in nM, among the highest, and concluded the opposite. Another entry in this challenge, NeuroChip Twin, found the trend; tracing that disagreement found our error.

On the ninth. A reviewer of the first round suggested that the twin raises firing rate under GABA-A block, where real chips lower it, because it lacks presynaptic vesicle depletion: in a dense culture, disinhibited bursts should exhaust the excitatory synapses and silence the network between them. It is a good hypothesis and the twin can test it, so we registered the test before running it (`preregistration/twin_depression_under_block.json`). We ran the four drug knobs and a five-step GABA-A dose series under each of the 7 depression settings of §13's earlier check. Depletion shrinks the rise, the more so the stronger it is, but in none of them does firing rate fall (0 settings), and in none do both burst rates fall with dose, as they do on real chips. The weights were not recalibrated with depression in place, so this does not rule the mechanism out; it does rule out depletion added to the present twin as the explanation.

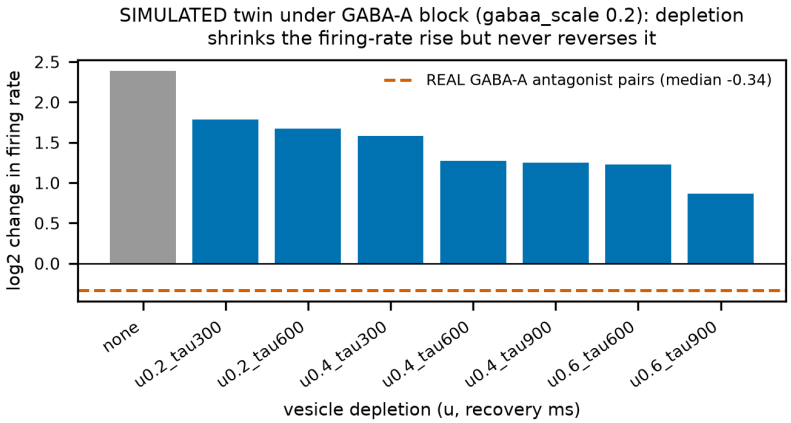


Figure 14. *Firing-rate change under strong GABA-A block in the SIMULATED twin, per vesicle-depletion setting, against the REAL median change on GABA-A antagonist pairs. Depletion shrinks the rise; no setting reverses it.*

19. What you cannot conclude from this work

- **No clinical or safety claims.** NeuroMPS is an exploratory analytical framework. We make no efficacy, diagnostic or clinical claims, and the design could not support them.
- **It is not a screening tool.** A within-platform mechanism accuracy of 0.537, with an interval from 0.4216 to 0.6845, and near-chance performance across laboratories do not make a decision instrument.
- **Interspecies support is narrow.** Rat-to-human transfer rests on 37 rat training pairs and 32 human testing pairs, which supports the direction of the effect rather than its size.
- **The simulator is not fully reconciled.** The twin sits inside the real band on two of five features, varies from seed to seed, and gets GABA-A potentiation mostly wrong.
- **The twin does not track dose.** Burst rates fall with bicuculline dose, but the twin's inferred inhibition does not, so its dose inference is not usable yet.
- **The species balance is severe.** Our asset holds 21,530 rat against 357 human records — a consequence of open-access licensing, not a design choice.
- **The samples are small.** Our primary classification rests on 73 paired recordings from 50 independent cultures.

ACT VI: Application value, translation and open verification

20. Application value and translation for neural organ-on-a-chip

The main bottleneck in making neural organ-on-a-chip systems useful is not microfluidic fabrication; it is making functional electrophysiology interpretable and reproducible. Academic centers and commercial vendors alike produce multi-channel MEA recordings that remain hard to compare across sites. Our results translate into practical guidance in five areas.

Where this sits. The US FDA Modernization Act 2.0 (2022) replaced the statutory reference to animal tests with "nonclinical tests", a category that names cell-based assays, organ chips and microphysiological systems, and computer modeling. That makes comparable, checkable readouts from neural chips more useful, and it is the setting our tools are built for. The provenance chain — license, source URL and a cryptographic checksum recorded for every file, and a build that halts on any mismatch — can serve as an open template for documenting where such data came from. It is a record of traceability, not a qualification process: NeuroMPS has not been qualified or reviewed by any regulator, and nothing in this report is evidence for a submission.

1. Design guidance for chip hardware. Developers often assume they can validate directional connectivity by measuring cross-correlation latencies or coupling asymmetries between open somatic chambers. §15 shows on real devices that this does not work: chamber readouts reached 0.3854 balanced accuracy where chance is 0.5, on a sample that could have detected 0.6438. To show that axons really conduct one way, you need microelectrodes inside the microchannels themselves — and knowing that saves time otherwise spent on measures that cannot answer the question.

2. A standardization framework for contract research. Screening protocols should move away from proprietary vendor summary tables toward standardized processing from raw spikes. Our feature ablation (§8) shows that two rate metrics match the predictive power of the full nine-feature vector, while avoiding the cross-platform variance that plagues burst-structure and synchrony metrics (§3, §11). A smaller, more reliable feature vector is the more reproducible instrument.

3. Cheaper assay design in silico. Plating human iPSCs is expensive and takes weeks. The twin gives you a testbed for pre-registering an assay: the power sweep in §17 sets a minimum electrode count (24 electrodes per compartment, 10-second recordings) for distinguishing wiring regimes, and the listening-radius finding in §13 gives you a way to diagnose electrode catchment overlap in physical recordings.

4. Paired-delta normalization for analysis pipelines. Resting firing rates in primary cultures span an order of magnitude. Expressing screening features as paired log-ratios against each well's own baseline stabilizes the analysis, raises posterior contraction from 0.3 to 0.5375, and removes most of the false-disinhibition artifact described in §16.

5. Sensitivity targets for an assay, set by a mapped brain (Act VII, added after the first round). An MEA assay is useful only if it flags a compound before the compound disturbs what a nervous system does. On a simulated fly brain, behaviour failed while the chip's firing rate had moved by as little as 0.149 log2 under GABA block. Two consequences follow for assay design, both from simulation and both to be checked on real tissue. First, measure each readout's untreated variation and state it: an assay meant to catch disinhibition in time needs firing-rate noise below that level, which the best real controls in our asset only just reach. Second, read a screen's hit threshold for what it can miss: the EPA's acute MEA screen calls a firing-rate hit only beyond a change of about −1.784 log2, 5.87 times the largest firing-rate change at which the fly's behaviour had already failed. That threshold is a reasonable guard against false hits; it is not evidence that a compound below it leaves circuits untouched. Third, where to record matters as much as how precisely: in the same simulation, electrodes on the brain's output tract caught GABA block in time where a planar array did not (§24), the same advice §15 gives from real devices: instrument the channels.

21. How to reproduce and check this work

Every computed number in this report carries an automated pointer back to the `outputs/metrics.json` key that produced it, and a test fails if any number drifts from that file. To reproduce everything from a clean machine, run these steps in a terminal:

step	command	what it does
1. get the code	`git clone https://github.com/Vodolay/AI4S`, then `cd AI4S`	downloads the repository
2. create the environment	install uv as the README shows, then `uv venv --python 3.11 .venv`	an isolated environment with the pinned Python
3. install	`uv pip install --python .venv/bin/python -e ".[dev]"`	NeuroMPS and its pinned dependencies
4. quick run	`.venv/bin/python run_all.py --quick`	every stage end to end on the reduced data tier
5. full run	`.venv/bin/python run_all.py`	the full data tier, 3,000 forward simulations, every figure
6. check the numbers	`.venv/bin/python -m neuromps.report.verify`	re-reads every citation in this report and fails on any disagreement
7. open the demo	`.venv/bin/streamlit run app/app.py`	interactive mechanism prediction, twin sweeps and record browser

Use the environment's own `python` and `streamlit`, as shown. We ran these steps on a fresh clone, and every cited number reproduced exactly except ten that depend on the machine: the timings, and the counts and intervals from posterior sampling that stops at a time limit (§16). Provenance is guarded throughout: `data/raw/MANIFEST.json` records every source's URL, license and checksum.

22. AI assistance disclosure

We built this project in a paired workflow with Anthropic's Claude Code. Claude wrote most of the code and tests under step-by-step human direction: we set the plan and the phase gates, chose the scope, approved each step, and directed the science. We also used it to draft and revise parts of this report, which we then edited ourselves. Three practices kept that collaboration honest. Numbers are emitted into `metrics.json` by scripts and re-checked by `neuromps.report.verify`, never typed by hand. We report results as we observed them, including the ones that failed. And `report/LAB_NOTEBOOK.md` records what we tried and abandoned. No AI model is part of the delivered pipeline itself: every statistical model in it is trained from scratch on the data described here.

ACT VII: From chip to animal (after round 1)

Added after the first-round deadline; `report/POST_SUBMISSION_CHANGES.md` records every change since the submission. Every result in this act is from simulation, for research decision support; nothing in it is a claim about any compound's effect in any animal or person.

23. The question a chip cannot answer on its own

A neural chip reports what a drug does to a dish of neurons. The question an assay is used to answer is different: will the drug disturb what a nervous system *does*? No culture can answer that, and so no chip assay can be checked against it. A mapped brain lets us ask. The adult fly brain has

been reconstructed synapse by synapse, and Shiu et al. (2024) turned that wiring diagram into a leaky integrate-and-fire model whose responses match the fly's own sensorimotor pathways.

We applied the twin's drug knobs to that model on the FlyWire connectome, 138,639 neurons joined by 15,091,983 connections, and read the same simulated brain two ways at once:

- **as a chip**: a virtual electrode array laid over the brain's regions, each electrode hearing one neuron, its recordings scored by `features/mea.py`, the extractor every real recording in the asset goes through;
- **as an animal**: the neurons the published model validated against behaviour. MN9, the proboscis motor neuron, under sugar drive (feeding) and again with bitter drive added (aversive gating); aBN1 under antennal mechanosensory drive (grooming); and how many descending neurons the sugar drive recruits (how selectively one stimulus reaches one output).

Five drug families were each run at four or five strengths: GABA block, block of the glutamate-gated chloride channel (GluCl), GABA potentiation, acetylcholine block and a raised spike threshold. In the fly, GABA and glutamate are the fast inhibitory transmitters and acetylcholine the fast excitatory one, so these are the fly's counterparts of the twin's GABA-A, AMPA and sodium-channel knobs (§14). Behaviour is summarised by a disruption index: the mean absolute log2 change of the four readouts.

How it was kept honest. The design, ten predictions and their pass/fail tests were committed before the simulations were run (`preregistration/fly_chip_to_animal.json`; its commit and checksum are in `fly_chip_to_animal.registration`), together with every exploratory result seen before registering. Two later addenda followed the same rule. The published neuron lists come from an older FlyWire release than the one we model, and a neuron's identifier changes when it is edited, so the code keeps only neurons whose identifier is unchanged and refuses any table that covers too few of the model's neurons; the two tables we use cover 0.9999 and 0.9964 of them. With the same code, sugar drives MN9 at 76.75 Hz on our release and 77.375 Hz on the paper's.

24. The chip ranks drugs correctly and raises the alarm late

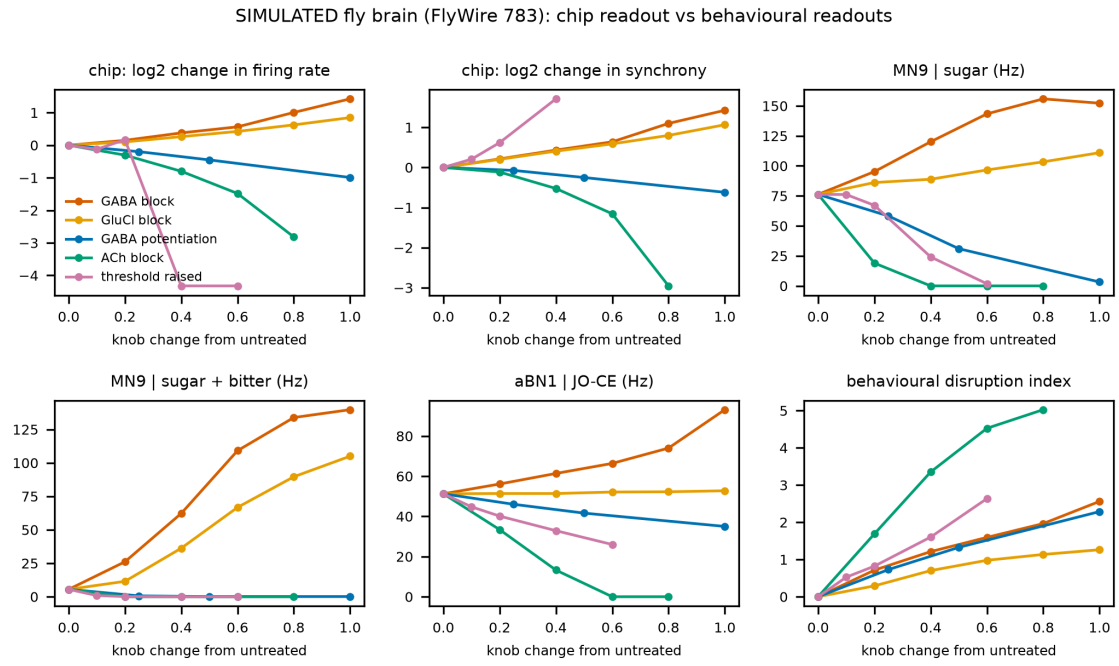


Figure 15. *Chip readouts and behaviour readouts of the SIMULATED fly brain against dose, one line per drug family. Top: the virtual array's firing-rate and synchrony changes, and MN9 under sugar drive. Bottom: MN9 with bitter drive added, aBN1 under antennal drive, and the disruption index.*

Behaviour responds as the circuits predict. Blocking GABA releases feeding from bitter suppression (rank correlation with dose 1.0) and recruits ever more descending neurons (1.0): the brain loses its selectivity. Blocking acetylcholine silences the motor and grooming outputs.

The chip ranks drugs by their effect on behaviour. Across the 21 drug conditions, the change in the array's firing rate tracks the disruption index (Spearman 0.863). The fly array also agrees with real chips about which way each mechanism moves each feature as often as the chip twin does: 12 agreeing features against the twin's 12.

But it is late: the registered prediction failed in every family. We predicted that, at the registered detection threshold, the chip would flag each drug at a dose no higher than the one that impairs behaviour. In all five families behaviour failed first. At the weakest GABA block, the bitter gate on feeding has already broken while the array's firing rate has barely moved. Dropping any one readout from the disruption index leaves the chip late in at least four of the five families, and the result replicates on fresh random seeds.

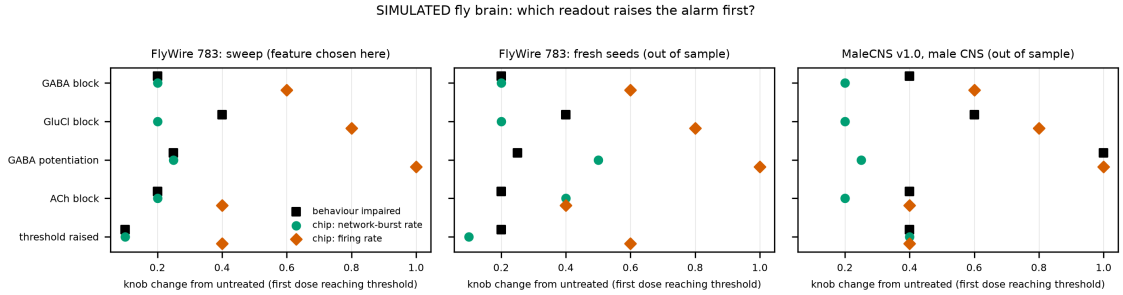


Figure 16. *Which readout raises the alarm first, SIMULATED. For each drug family: the first dose that impairs behaviour (black), and the first at which the array's network-burst rate (green) and firing rate (orange) change by the registered threshold. Left: the sweep in which network-burst rate was picked out; middle and right: the two out-of-sample runs. Network-burst rate's early crossings are within untreated noise (see text).*

What a chip would need to catch it in time. The change in firing rate at the first dose that impairs behaviour is the precision a chip must reach to flag that drug before behaviour fails. This analysis is exploratory: it was done after the registered test failed.

Drug family	Firing-rate change at behavioural onset (log2)
GABA block	0.149
Raised threshold	0.124
GABA potentiation	0.199
GluCl block	0.263
Acetylcholine block	0.304

Real chips in the asset vary between untreated recordings by up to 0.156 (Kapucu's vehicle controls, 95th percentile) and 1.752 (Vassallo's inactive compounds). At the better of the two, a chip would flag acetylcholine block, GluCl block and GABA potentiation in time, but not GABA block or a raised threshold; at the worse, none of the five. GABA block was the hardest family to catch in the second brain as well. The practical reading is a specification, not a verdict: an assay meant to catch disinhibition before it reaches behaviour must resolve firing-rate changes of a few percent.

A tempting fix that is noise. In the first sweep, network-burst rate crossed the threshold at or before behaviour in every family. We registered a confirmation, on fresh seeds and on the second brain, before seeing either. Out of sample it crossed early in 8 of 10 family tests against 3 for firing rate, and two of the four registered tests passed. Then we checked what the registration had not: two groups of *untreated* recordings already differ by the threshold in a fraction 0.382 of random splits, while for firing rate the untreated 95th percentile is 0.028 log2. Trains shifted in time, which keeps every rate and breaks every coincidence, still yield network bursts. Network-burst rate looked early because it is noisy, not because it sees more.

Where to put the electrode. The fly's descending neurons carry the brain's output down the neck to the nerve cord: a bundle of axons between two compartments, which is what a chip's microchannels are. We registered a test before recording them (`preregistration/fly_output_tract_electrode.json`): read the same spontaneous recordings through 64 single-unit electrodes on descending neurons and through 64 electrodes drawn from the planar array, at one threshold set by real-chip noise. All four registered predictions passed. Under GABA block, at the dose where behaviour first fails, the tract's firing rate has moved by 0.361 log2 against 0.044 for the equal-count planar array and 0.149 for the full one. The tract flags 3 of the five families no later than behaviour, against 2, and never false-alarms on untreated recordings (rate 0.0). Two checks we added after it passed temper it without overturning it. Which 64 neurons the electrodes hear matters: over ten random draws of each array, the tract is ahead in 7 of the 10 draws. And the result does not come from the overlap between the tract and one behaviour readout, which counts descending neurons: leaving that readout out changes nothing. This agrees with what real devices showed in §15, from the other side: there, chamber electrodes could not tell which way axons fired and only electrodes inside the microchannels could; here, an electrode on the output tract sees the circuit fail sooner than one on the tissue around it.

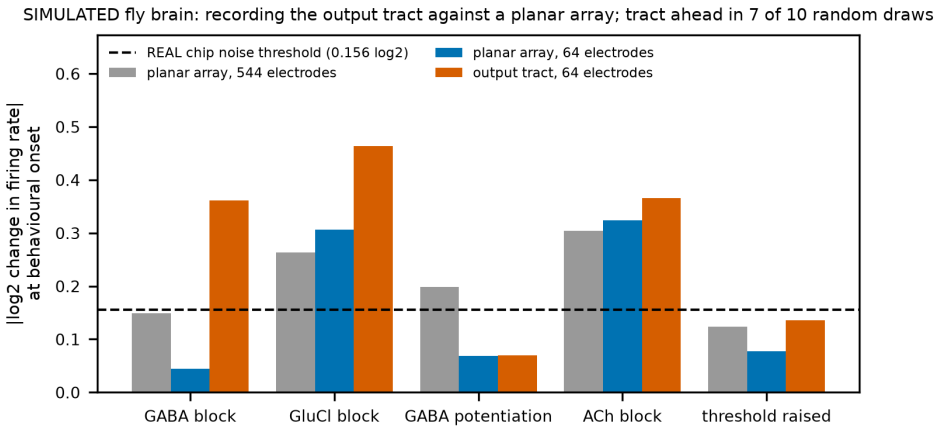


Figure 17. *Firing-rate change at behavioural onset in the SIMULATED fly brain, per drug family, read through the full planar array, 64 of its electrodes, and 64 electrodes on the descending neurons of the output tract, against a threshold set by REAL chip noise.*

25. Direction between regions, and a second brain

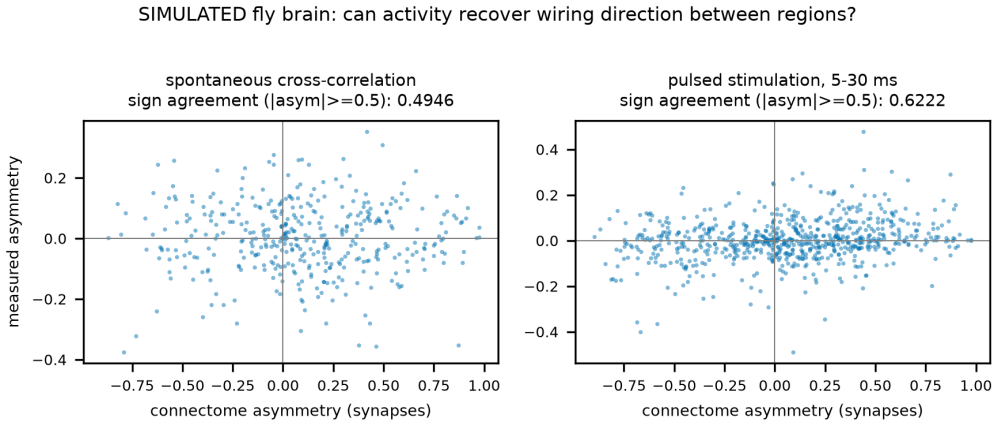


Figure 18. *Region-to-region direction in the SIMULATED fly brain. Each point is a pair of brain regions joined by many synapses; the horizontal axis is the connectome's true asymmetry, the vertical the measured one. Left: the chip's cross-correlation measure on spontaneous activity. Right: pulsed stimulation of one region, read shortly after each pulse.*

§15 and §17 found that a chamber-level measure does not recover direction on real devices whose axons are instrumented. The fly brain repeats that test where the answer is known for every pair of regions. On spontaneous activity, the chip's cross-correlation measure gets the direction of strongly

asymmetric pairs right at chance (0.4946). Pulsed stimulation of one region, read shortly after each pulse, does better (0.6222) but misses the registered target, and read through the array's electrodes alone it falls back to chance (0.4889).

A second brain agrees. The male central nervous system connectome (MaleCNS: brain and nerve cord, an independent reconstruction of a male fly; 165,122 neurons and 25,563,096 connections) was run under the same design, with the published neuron sets matched by cell type and side. All five registered replication tests pass: sugar drives MN9 at 31.625 Hz and bitter suppresses it; GABA block releases it; acetylcholine block silences the outputs; and the chip's firing-rate change again tracks behavioural disruption (Spearman 0.878).

One step closer to behaviour: motor neurons. The behaviour readouts above are named interneurons, MN9 excepted. The male connectome includes the nerve cord, so the same protocols can be read at the motor neurons that move the body: the 67 proboscis motor neurons for feeding and its bitter gate, the 135 front-leg motor neurons for antennal grooming, and how many of all 815 motor neurons a drive recruits. We registered the test before running it (`preregistration/fly_motor_readout.json`), repeating the replication's trials with the same seeds, so its interneuron readouts had to come back unchanged; they did (largest difference 0.0). The pools carry the behaviours: sugar drives the proboscis pool at 31.405 Hz and bitter cuts it to 5.625 Hz; antennal drive moves the front legs at 5.023 Hz against 2.394 under sugar. Six of the seven registered predictions passed: GABA block releases the gate at the motor level, acetylcholine block silences motor output, and the chip's firing-rate change tracks motor disruption (Spearman 0.8922). The one that failed was the chip-is-late prediction, and narrowly: the chip flagged the drug after motor output was impaired in 3 families and at the same dose in 2, where the test needed four. It was never earlier (0 families). Motor output failed no later than the interneuron readouts in every family, and earlier under acetylcholine block. Read at the motor level, then, the chip is not ahead of behaviour, it is at best level with it.

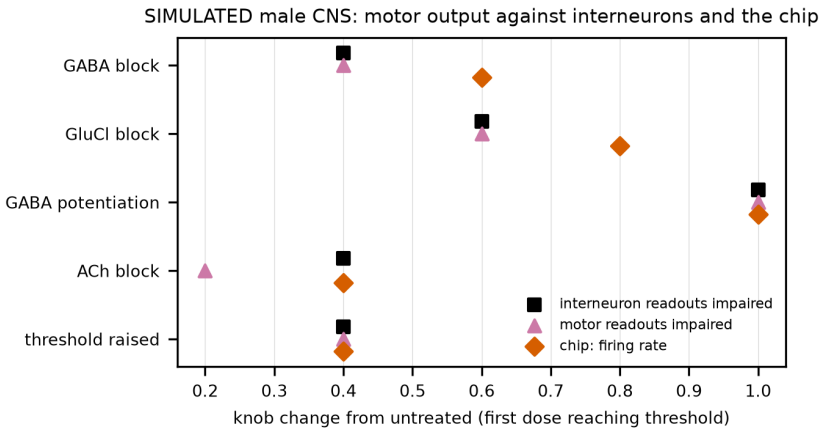


Figure 19. *First impairing dose at the interneuron and motor levels, and first chip detection, per drug family, in the SIMULATED male CNS. The chip is never earlier than motor output.*

The two ties were a matter of resolution: both onsets fell inside one coarse dose step. So we registered a follow-up before running it (`preregistration/fly_motor_tie_break.json`), with four finer levels inside each tied step, the same seeds and the same untreated references; re-running the tied levels reproduced them exactly (largest difference 0.0). The raised threshold resolves: motor output fails at 0.32, the chip detects at 0.4. GABA potentiation stays tied even on the finer grid, both crossing at 0.6; there the chip is within a hair of its threshold one step earlier. The registered follow-up predicted that both ties would resolve with the chip later, so it failed too. MR5 stays failed as registered; read with the follow-up beside it, the chip is later than motor output in 4 families, level in 1 and earlier in 0.

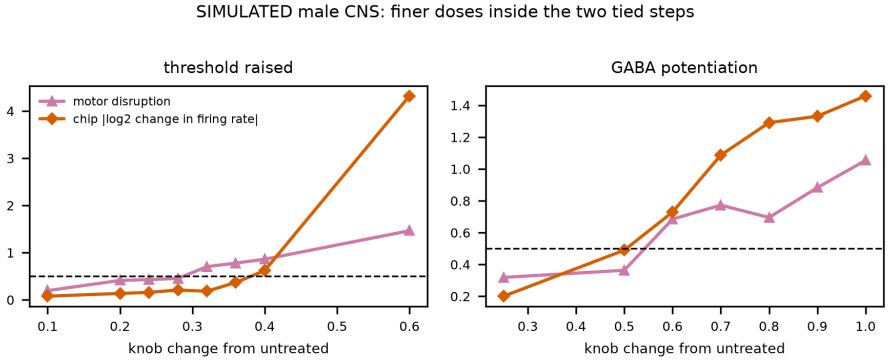


Figure 20. *Chip firing-rate change and motor disruption across the two tied dose steps, on a finer grid, in the SIMULATED male CNS; dashed line, the 0.5 threshold both are read against.*

26. What the fly study does not show

- **It is a model.** The fly model is a leaky integrate-and-fire network with one free weight, no adaptation and no neuromodulation; it fires asynchronously and does not burst the way a culture does. Its behaviour readouts are firing rates of named neurons, motor neurons included, not movement: there is no body and no sensory feedback. In the male connectome, which has more synapses per neuron under the same weight, any drive recruits a large share of the motor neurons, so off-target recruitment starts high.
- **The knobs scale whole transmitter classes.** A real compound acts on particular receptor subunits, with its own potency and off-target effects. Nothing here says what any compound does to a fly, a culture or a person.
- **The precision table is exploratory.** It depends on the disruption index and the registered threshold, and it was computed after the registered test failed.
- **The real-chip noise floors** come from two sources with very different spreads, one of them from few recordings.

The study runs with `python run_all.py --stage fly` (the licensed files are fetched first); `python -m neuromps.fly.replication`, `python -m neuromps.fly.confirm` and `python -m neuromps.fly.checks` run the two addenda and the noise checks. The demo app's fourth tab shows the sweep with a dose slider.

ACT VIII: The embodied fly (after round 1)

Act VII read behaviour as the firing of named neurons. This act gives the simulated fly a body and asks whether Act VII's answer survives: does the chip still flag a drug before behaviour fails when behaviour is movement? Every stage below was registered before it ran; the roadmap document (`report/ROADMAP_EMBODIED_FLY.md`, §7) carries each stage in full, including the failures, and `report/DECISION_LOG.md` records every choice of path. Everything here is simulated.

27. A body for the simulated fly

The male nerve cord (male CNS connectome) drives a muscle-driven front leg (FlyMimic, in MuJoCo) through a motor-neuron-to-muscle map built from published names; the FlyWire brain drives the proboscis of a whole-fly model (flybody) through the motor neurons of each proboscis muscle. The two connectomes are joined where each model is validated: the FlyWire brain's descending spikes are replayed into the male nerve cord, matched by cell type and side. Antennal input to the brain then brings the claw to 0.5369 mm of the antenna, sugar input only to 0.9377 mm, and removing the two antennal-grooming descending neuron types alone leaves it at 1.4793 mm. Sugar input extends the proboscis fully (0.3431 mm of a possible 0.3428), and adding bitter input halves the peak (0.1618 mm).

28. Walking: what the wiring does and does not do

The male nerve cord, run as leaky integrate-and-fire neurons, does not turn its walking command into a rhythm: not as published, not with spike-frequency adaptation swept over a registered grid, not with either of two rules for where glutamate excites. A firing-rate model published for this question (Pugliese et al. 2025), reimplemented and built on the male CNS as its authors build it, does produce a front-leg motor rhythm under the walking command: a corrected rhythm score of 0.8886 at a period of 196.5 ms without size scaling, and silencing the published three-neuron circuit removes it (0.0). It depends on how neuron size is treated, and neither model alternates antagonist muscles. One earlier claim was withdrawn on the way: the rhythm measure registered since the first leg stage took the largest autocorrelation without requiring a peak, so smooth movements scored as rhythmic; a registered audit with a corrected measure reproduced every recorded outcome and reversed one (that walking drive "swings" the leg).

29. Closing the loop

Feedback from the front leg's position sensors (the femoral chordotonal claw neurons, identified by type in the male CNS) was closed through the leg every millisecond. It shifts the tibia motor pools toward resisting the movement, but it neither moves a held leg nor returns a pushed one faster: the push returns in 32.0 ms with feedback and 30.0 ms without. The male CNS traces few proprioceptors for the front leg (4 claw neurons on the left, against 22 on the left hind leg), which is the most likely limit. Closed through the well-traced right hind leg instead (25 claw neurons; flybody's servoed tibia), the feedback changes the leg's posture but makes the hold swing, and a registered paired test on fresh seeds found no part of it that speeds the return: resistance sooner than a sham in 14 of 32 seeds.

30. Drugs in the loop: the chip is late at every level

Act VII's drug design, rerun with movement as the readout (proboscis extension and the grooming reach), gives the same answer. In every family the chip first flags the drug at a higher dose than movement fails: GABA-A block impairs movement at 0.2 of the knob's range and the chip flags it at 0.6; GABA-A potentiation impairs movement at 0.5 and is flagged at 1.0. Movement never fails before the neural readouts of Act VII, and the output-tract electrode no longer leads the planar array at this level. A registered rerun on fresh seeds, against the chip's own fresh-seed replication, finds the chip late again in every family. Left out of the index, the bitter-gated proboscis turns out to carry GABA-A potentiation: without it movement fails only at 1.0, where the chip flags it. At the dose where movement first fails the chip has moved by only 0.149 to 0.452 log2 (exploratory), within the noise of real chips. A literature comparison registered before its search found no study that gives adult flies these drugs and scores feeding or grooming movements, so the simulated directions cannot yet be checked against fly pharmacology: the gap names the experiments that would. Simulating the identified neurons such a study manipulated (the front leg's GABAergic 13A and 13B hemilineages) reproduced one published direction for the grooming reach: silencing 13A shortens it to 0.3806 mm. Read at the femur-tibia joint, though, none of the published postures appears (silencing 13A shifts the tibia by -0.1488 rad, toward extension, against a published

flexion): in this model the inhibitory leg neurons act as broad brakes, not as joint-specific controllers.

31. What the embodied fly does not show

- **It is open loop where it works and weak where it is closed.** The leg and proboscis results are open loop; the one closed loop runs through four sensory neurons.
- **The bodies are partial.** One muscle-driven leg; the proboscis joints are position servos.
- **The knobs are transmitter-class scalings,** as in Act VII; nothing here says what any compound does to a fly or a person.
- **The walking result is a negative for one model class** and a reimplementation of another group's model; neither produces coordinated walking.

The stages run with `python run_all.py --stage embodied` (or one at a time through `python -m neuromps.fly.pipeline`).

Appendices: consolidated technical reference

Appendix A & B — Record architecture, data levels and ingestion ETL

Every record declares a data level: `spike_times`, `per_well_features` or `fitted_curve`. The level decides what may be computed from the record, and the build enforces that instead of trusting it — a feature that needs spikes cannot be derived from a record holding only a fitted curve. Records validate against `schema/neuromps_schema.json` at three conformance levels, so a partially specified source can still enter the asset as long as it declares what it is missing.

We record quality-control flags but never apply them silently: 537 of 21,887 records are marked unusable, with the diagnostic reason attached, and the flag travels with the record. Each analysis decides whether to honor it and says which it did; if we dropped flagged records inside the pipeline, the reader could never see the drop.

Each record also carries the method string that produced its features — the exact burst rule, the synchrony method and the active-electrode threshold — so two records computed under different settings cannot be compared by accident.

Ingestion lives in `src/neuromps/fetch/`. It uses HTTP range requests against remote ZIP archives to pull only the files we need, instead of downloading multi-gigabyte archives. Two server behaviors needed special handling and now have regression tests: one EPA host returns HTTP 500 for suffix ranges, and the same host, when offered gzip encoding, replies with a plausible `Content-Range` while actually streaming the whole 7.5 GB file from byte zero.

Appendix C — Data sources and licenses

dataset	platform format	species	experimental role	records	status
`toxcast_mea`	screening MEA	rat	high-throughput reference, fitted curves	20,375	ingested
`screenneuropharm_bic`	two-compartment chip	rat	directionality and twin calibration	580	ingested
`kapucu2022`	planar 48-well MEA	rat / human	primary ML benchmark and transfer	288	ingested
`vassallo2017`	planar MEA	rat	second within-platform benchmark	283	ingested
`trujillo2019`	planar MEA	human organoid	baseline electrophysiology	184	ingested
`mateus2024`	microchannel MEA	rat	microchannel propagation asymmetry	108	ingested
`brewer2021`	microchannel MEA	rat	axonal propagation ground truth	40	ingested
`sharf2022`	high-density MEA	human organoid	negative-control screening (diazepam)	29	ingested
`fofie2025`	multi-well MEA	human sensory	rate-only validation arm (§8)	291 pairs	rate-only, outside the record asset
`tukker2020`	planar MEA	rat / human	neurotoxicity screening	—	**excluded, CC BY-NC**

`fofie2025` reports total spike counts rather than spike times, so it supports firing-rate features only and is not part of the 21,887-record asset; we use it solely for the cross-source rate-only arm. We record `tukker2020` as a skipped source, with its reason, rather than silently leaving it out.

Appendix D & E — The nine-feature vector, and enforced repository invariants

feature	formal definition as computed in `features/mea.py`
`mfr_hz`	spikes per second per active electrode
`burst_rate_per_min`	bursts per minute per active electrode (≥ 5 spikes, ISI < 100 ms)
`burst_duration_s`	mean duration from first to final spike across identified bursts
`spikes_per_burst`	mean count of spikes participating in a burst
`pct_spikes_in_bursts`	share of all spikes occurring inside bursts
`network_burst_rate_per_min`	synchronous population events per minute, clearing chance participation
`synchrony_index`	mean pairwise spike-time tiling coefficient, coincidence window Δt = 50 ms
`isi_cv`	coefficient of variation of inter-spike intervals
`active_electrodes`	electrodes clearing 5 spikes per minute

Each of the following is a test that fails, not a convention we merely follow.

- **Provenance.** The asset build fails if any record's provenance hash disagrees with the manifest.
- **One feature extractor.** The twin computes features through the same `features/mea.py` as the real recordings and — after a defect that survived several phases — through the same coupling measure on the same per-electrode input.
- **Interval discipline.** A contract test fails if any headline metric loses its confidence interval, and the resample count sits next to every estimate.
- **Cache meaning.** The simulation cache stores its summary names, electrode count, a hash of the code that defines every summary, and the calibration it was generated under; it refuses to load across a change in any of them.
- **Report numbers.** `neuromps.report.verify` re-reads every citation in this report and fails on any disagreement.
- **No clinical claims.** A test scans the reports for efficacy, diagnostic and treatment language and fails unless the line is a disclaimer ruling it out.

Appendix F & G — Coupling asymmetry proof, and posterior calibration diagnostics

Appendix F. The measure the twin exists to use reports relative firing rate as much as pathway direction. The cleanest demonstration holds the wiring **fixed** and changes only the drive: lowering the receiving chamber's background drive moves asymmetry from 0.109 to 0.681 with the pathway unchanged. Across the whole sweep, drive moves the measure further than wiring does: a span of 0.671 against 0.109. Until something rules it out, an open-chamber asymmetry is a statement about which chamber is busier.

Appendix G. We trained on 3,000 simulations at the calibrated density, of which 0 produced no spikes. The two parameters that fail rank uniformity are the connectivity parameter and the inhibitory knob — the latter being the one §16 depends on, which is why we do not quote its intervals as calibrated.

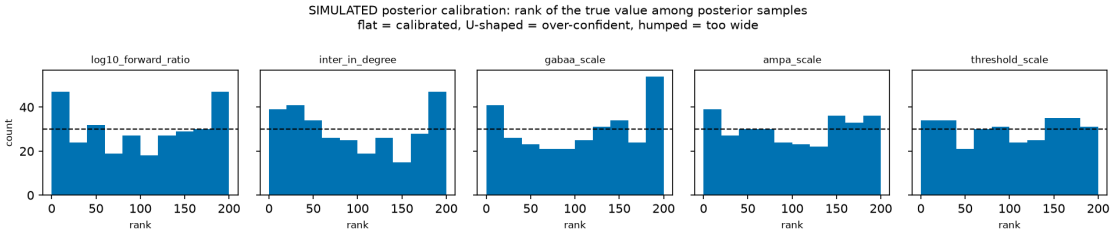


Figure 21. *SIMULATED posterior calibration: the rank of the true parameter among posterior samples. Flat would be calibrated; these are not.*

Appendix H — Quality-control flag registry

QC flag	affected records	identification criteria	analytical handling
`incomplete_dose_series`	272	concentration series missing intermediate dilution steps	flagged; excluded from monotonic dose sweeps
`too_few_active_electrodes`	265	fewer than 10 electrodes clearing 5 spikes/min	flagged; preserved in the data asset
`replicate_divergence`	221	well deviates more than 3σ from its culture replicate cohort	flagged; logged in metadata
`outlier_feature_value`	221	feature value outside the acceptable biological distribution	flagged; excluded from density fitting
`excluded_well_upstream`	90	source laboratory marked the well defective	flagged; excluded from benchmarks
`missing_baseline_pair`	24	no baseline recording on the same well	flagged; cannot enter a paired-delta model
`noisy_electrodes_present`	11	significant mechanical or electrical artifact	flagged; isolated during ingestion
`unreadable_channels`	2	channel could not be decoded from the deposit	flagged; channel dropped

Appendix I — Digital twin forward-model parameters

Calibrated values are the chosen draw from the §13 search; held values were fixed and not searched.

parameter	role	calibrated value	how it was set
`n_per_compartment`	plating cell density per chamber	967 neurons	calibrated, searched over 250–1500
`detection_radius_um`	virtual electrode listening catchment	92.5135 µm	calibrated, searched over 40–100 µm, capped at half the electrode pitch
`intra_in_degree`	recurrent local synaptic inputs	18.4929 per neuron	calibrated, searched over 10–35
`background_rate_hz`	Poisson drive frequency	3.3596 Hz	calibrated, searched over 1–40 Hz
`background_weight`	Poisson drive conductance step	0.0692	calibrated, searched over 0.005–0.12
`excitatory_weight`	excitatory conductance step	0.386	calibrated, searched over 0.15–0.45
`inhibitory_weight`	inhibitory conductance step	5.4261	calibrated, searched over 1–7
`adaptation_increment`	spike-frequency adaptation step	1.0093	calibrated, searched over 0.5–2.5
`threshold_spread`	per-neuron excitability heterogeneity	0.2258	calibrated, searched over 0–0.3
`adaptation_tau_ms`	adaptation decay time constant	380 ms	**held**, literature standard
`intra_delay_sd_ms`	conduction delay dispersion	0.0 ms	**held**, one delay for every synapse
`connection_sigma_um`	range of the within-compartment wiring rule	unset	**held**, uniform random wiring
`inter_in_degree`, `forward_ratio`, drug knobs	cross-chamber connectivity, pathway asymmetry, pharmacology	—	**inferred** by simulation-based inference (§16)

All conductance steps are dimensionless in this model; they are not membrane voltages.

Appendix J & K — Microchannel detector robustness, and timings

Appendix J. Counting recordings, median absolute asymmetry runs from 0.3392 for control geometries to 0.8543 for the "rams" geometry. Counting biological chips instead, it is 0.6168 against 0.8745, a difference the test does not resolve: Kruskal p across the geometries is 0.63969, and every Holm-adjusted comparison against the control geometry is 1.0. The null is not an artifact of our geometry assumptions either: it survives every detector variant we tried, with a worst-case Kruskal p of 0.13074. Tolerating single-electrode dropouts triples what the detector has to work with, from 336,794 analyzed events to 1,101,233, and does not change the outcome. We also measured and report a counting confound: chips with fewer propagation events look more one-sided (Spearman ρ = −0.494, p = 0.01032), which is what a counting artifact looks like. Condition is also confounded with culture batch by the deposit's own design.

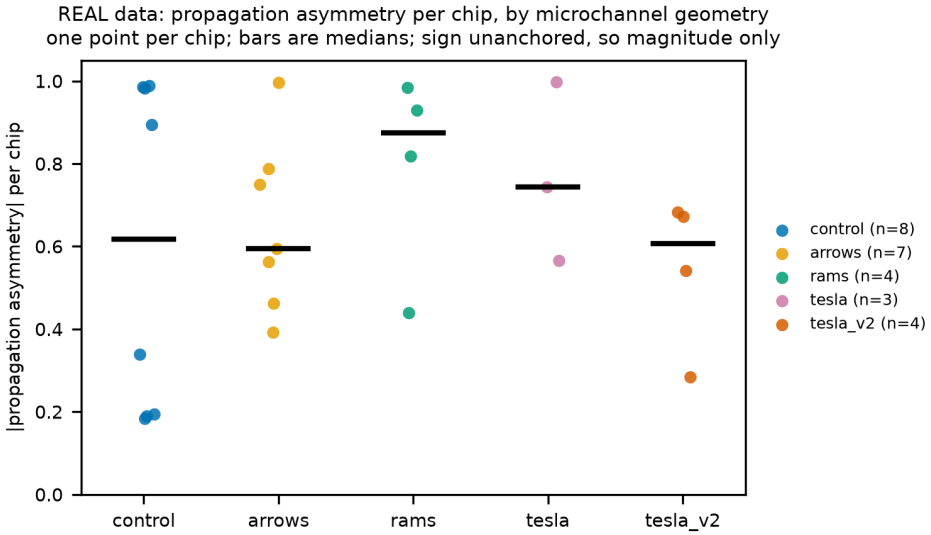


Figure 22. *Microchannel propagation asymmetry by device geometry, REAL recordings.*

Appendix K. `--quick` completes end to end in 544.7 s on a workstation CPU with 20 cores, and in under twenty minutes on a dual-vCPU CI runner; the two differ because `worker_processes` correctly asks for the cores it is given. The run itself writes that figure, and only a run that executed every stage can write it, so a partial re-run cannot quietly shrink it. It is one measurement rather than a bound: repeat runs on the same machine vary with whatever else the machine is doing, and the slowest we have seen crossed ten minutes. The longest stage is the twin, at 344.3 s. Full reproduction runs through `python run_all.py` after a full data-tier fetch. The environment is pinned in `pyproject.toml` and recorded from what actually imported: Python `3.11.16`, NumPy `2.3.5`, Brian2 `2.9.0`, PyTorch `2.14.0` and sbi `0.27.0`. We hold NumPy back because its next minor release removes `ndarray.ptp`, which Brian2 imports.

Appendix L, M & N — Sensory validation, dose tracking and species transfer

Appendix L, the rate-only arm. Human sensory neurons across 291 pairs in `fofie2025` shift by −0.1743 log2 units for vehicle controls, against −1.9529 for sodium-channel blockers, which yields 0.5012 balanced accuracy against a chance of 0.3333.

Appendix M, Task 2 dose tracking. The detection rate rises from 0.179 at 0.1 µM to 0.426 at 30 µM. Quoted on its own, that would look like a dose response, so here is the rest of the series: the rate peaks at 0.48 at 100 µM and falls to 0.091 at the top of it, 500 µM. The overall rank correlation is not significant (ρ = −0.059, p = 0.189).

Appendix N, species transfer, all settings.

transfer setting	balanced accuracy	95% bootstrap CI
human within-species	0.646	0.483 to 0.78
rat within-species	0.543	0.458 to 0.7
rat → human, zero-shot	0.55	0.4 to 0.6
rat → human, CORAL	0.442	0.302 to 0.591
rat → human, standardized	0.408	0.267 to 0.568

Every interval resamples cultures, not rows. For the within-species rows it is taken over the out-of-fold predictions pooled across the repeated grouped splits, and a culture drawn into a resample brings all of its predictions with it; for the transfer rows it is taken over the human test cultures.

Appendix O — Third-party software, models and AI tools

Software. We use the libraries below unmodified, as dependencies; none of their source is redistributed here. Exact versions are pinned in `pyproject.toml`, and `THIRD_PARTY.md` lists each package's license as read from its installed metadata.

license	packages	what they do here
BSD (3-clause and BSD-style)	numpy, scipy, pandas, scikit-learn, h5py, altair, xlrd	numerics, data handling, classifiers, charts, file readers
MIT	jsonschema, openpyxl, python-docx, pytest, ruff	schema validation, spreadsheet input, report build, tests, linting
Apache 2.0	sbi, pyarrow, requests, streamlit	posterior estimation, storage, downloads, the demo app
Apache and BSD (combined notice)	torch	neural network backend for posterior estimation
Python Software Foundation	matplotlib	figures
CeCILL 2.1	brian2	spiking-network simulation of the twin

Brian2's license is the most restrictive in the list; we use it unmodified, as a library.

Models. We use no pretrained models. Every statistical model in this report — logistic regression, gradient boosting and the neural posterior estimator — is trained from scratch on the data described here. The fly study of Act VII adopts the equations and constants of the whole-brain model of Shiu et al. (2024; code under the MIT license), reimplemented in our own code.

Data. Every dataset we used, with its license and status, is listed in Appendix C.

AI tools. We developed the project with Anthropic's Claude Code (§22), a commercial tool used under Anthropic's terms of service. It is not part of the delivered pipeline, and nothing it wrote is accepted without passing the same tests and citation checks as the rest of the code.